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VESALIUS.Geo
Web Services

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1 Concept

VESALIUS.Geo being an Enterprise Solution, It is required for external systems to communicate with data based on real time access.

This is facilitated by the standard set of available Web Services using industry standards. Details of all Web Services are explained below based on respective group.

2 Authentication

2.1 Authentication

Login			
Description:	To authenticate the user session by issuing a web session id. Once this API is called, the system returns a session id which will be used to validate the users session.		
Input Parameters	**Description**	**Type**	**Mandatory**
COMPANY_CODE	Company code	String	Y
SYSTEM_CODE	System Code	String	Y
PASSWORD	Password	String	Y
Success Output:			

```
<?xml version="1.0" encoding="UTF-8">
<Result>
  <Token>
    <Token_number>250410461752</Token_ number>
  </Token>
</Result>
```

Messages Used

Message Code	Message Type	Message
WS-00001	ERROR	ORGANIZATION INFORMATION IS MANDATORY.
WS-00002	ERROR	CODE INFORMATION IS MANDATORY.
WS-00003	ERROR	PASSWORD INFORMATION IS MANDATORY.
WS-00004	ERROR	ORGANIZATION DOES NOT EXIST IN THE SYSTEM PLEASE CHECK.
WS-00005	ERROR	AUTHENTICATION FAILED.
WS-00006	ERROR	MAXIMUM NUMBER OF CONNECTIONS REACHED.
WS-00007	ERROR	DATABASE ERROR.

2.2 Logout

Logout			
Description:	To end the user session.		
Input Parameters	**Description**	**Type**	**Mandatory**
TOKEN_NUMBER	Web Session ID – Used to check for active sessions only	Numeric	Y
Success Output:			

```
<?xml version="1.0" encoding="UTF-8">
<Result>
  <Success>
   <SuccessMessage> SUCCESSFULLY LOGGED OUT </SuccessMessage>
  </Success>
</Result>
```

Messages Used

Message Code	Message Type	Message
WS-00039	SUCCESS	SUCCESSFULLY LOGGED OUT.
WS-00008	ERROR	TOKEN NUMBER IS MANDATORY.
WS-00009	ERROR	TOKEN NUMBER IS INVALID.
WS-00007	ERROR	DATABASE ERROR.
WS-00010	ERROR	TOKEN NUMBER HAS BEEN ALREADY LOGGED OUT.
WS-00011	ERROR	TOKEN NUMBER DOES NOT EXIST IN THE SYSTEM.

2.3 GetUserGroups

GetUserGroups			
Description:	To end the user session.		
Input Parameters	**Description**	**Type**	**Mandatory**
USER_NAME	Valid VESALIUS User Name, For which request is being sent	String	Y
COMPANY_CODE	Company Code (Hospital Code)	String	Y
TOKEN_NUMBER	Web Session ID – Used to check for active sessions only	Numeric	Y

Success Output:

```
<?xml version="1.0" encoding="UTF-8"?>
<Result>
  <User>JANEFOO</User>
  <Group>
    <Code>34456</Code>
    <Name>Admin</Name>
  </Group>
  <Group>
    <Code>952627</Code>
    <Name>HIS Admin</Name>
  </Group>
  <Group>
    <Code>723728</Code>
    <Name>Clinical Admin</Name>
  </Group>
  <Group>
    <Code>526728</Code>
    <Name>BIlling and BO</Name>
  </Group>
</Result>
```

Error Output :

```
<?xml version="1.0" encoding="UTF-8"?>
<Result>
  <Error>
    <ErrorCode>NH-00215</ErrorCode>
    <ErrorMessage>NO EXISTING USER.</ErrorMessage>
  </Error>
</Result>
```

Messages Used

Message Code	Message Type	Message
WS-00001	ERROR	ORGANIZATION INFORMATION IS MANDATORY.

WS-00004	ERROR	ORGANIZATION DOES NOT EXIST IN THE SYSTEM PLEASE CHECK.
WS-00005	ERROR	AUTHENTICATION FAILED.
WS-00039	SUCCESS	SUCCESSFULLY LOGGED OUT.
WS-00008	ERROR	TOKEN NUMBER IS MANDATORY.
WS-00009	ERROR	TOKEN NUMBER IS INVALID.
WS-00010	ERROR	TOKEN NUMBER HAS BEEN ALREADY LOGGED OUT.
WS-00011	ERROR	TOKEN NUMBER DOES NOT EXIST IN THE SYSTEM.
WS-00013	ERROR	AUTHENTICATION FAILED PASSED TOKEN DOES NOT BELONG TO THIS ORGANIZATION.
WS-00042	ERROR	USER NAME NOT PASSED
WS-00044	ERROR	USER IS INVALID IN THIS ORGRANISATION

2.4 CheckActiveUserSession

CheckActiveUserSession			
Description:	Check Active User Session - External Reports.		
Input Parameters	**Description**	**Type**	**Mandatory**
TOKEN_NUMBER	Web Session ID – Used to check for active sessions only	Numeric	Y
USER_NAME	Valid VESALIUS User Name.	String	Y
User Session ID	Valid encoded VESALIUS login session ID	String	Y
Output: This session is Still Valid in VESALIUS			
<?xml version="1.0" encoding="UTF-8"?> <Result> <Active>Y</Active> </Result>			
Output : Passed Session is not valid in VESALIUS			
<?xml version="1.0" encoding="UTF-8"?> <Result> <Active>N</Active> </Result>			

Messages Used

Message Code	Message Type	Message
WS-00001	ERROR	ORGANIZATION INFORMATION IS MANDATORY.
WS-00004	ERROR	ORGANIZATION DOES NOT EXIST IN THE SYSTEM PLEASE CHECK.

WS-00005	ERROR	AUTHENTICATION FAILED.
WS-00039	SUCCESS	SUCCESSFULLY LOGGED OUT.
WS-00008	ERROR	TOKEN NUMBER IS MANDATORY.
WS-00009	ERROR	TOKEN NUMBER IS INVALID.
WS-00010	ERROR	TOKEN NUMBER HAS BEEN ALREADY LOGGED OUT.
WS-00011	ERROR	TOKEN NUMBER DOES NOT EXIST IN THE SYSTEM.
WS-00013	ERROR	AUTHENTICATION FAILED PASSED TOKEN DOES NOT BELONG TO THIS ORGANIZATION.
WS-00042	ERROR	USER NAME NOT PASSED
WS-00044	ERROR	USER IS INVALID IN THIS ORGRANISATION

3 Patient

3.1 GetPatientData

GetPatientData			
Description:	Get patient's personal data.		
Input Parameters	**Description**	**Type**	**Mandatory**
PRN	PRN/Document Number of a Patient	String	Y

Sample Output:

```
<?xml version="1.0" encoding="UTF-8"?>
<Result>
  <Patient>
    <PRN>2014-00001234</PRN>
    <Name>
      <Title>MS</Title>
      <FirstName>Alexandra</FirstName>
      <MiddleName></MiddleName>
      <LastName>Pheo</LastName>
    </Name>
    <Resident>Y</Resident>
    <DOB>12-Aug-1945</DOB>
    <Sex>
      <Code>F</Code>
      <Description>Female</Description>
    </Sex>
    <HomeAddress>
      <Address1>NO 18 TAMAN PUTERI JAYA</Address1>
      <Address2>JALAN 12</Address2>
      <Address3></Address3>
```

```
      <CityState>CHERAS SELANGOR</CityState>
      <PostalCode>43000</PostalCode>
    </HomeAddress>
    <ContactNumber>+65905412</ContactNumber>
    <Document>
      <Code>INTERFACE</Code>
      <Description>Interface PRN</Description>
      <Value>2014-00001234</Value>
      <ExpireDate></ExpireDate>
    </Document>
    <Document>
      <Code>PASSPORT</Code>
      <Description>Passport</Description>
      <Value>E78877998</Value>
      <ExpireDate>24-Dec-2013</ExpireDate>
    </Document>
    <Nationality>
      <Code>SG</Code>
      <Description>Singaporean</Description>
    </Nationality>
    <ChargeCategory>
      <Code>05</Code>
      <Description>NTUC MEMBER</Description>
    </ChargeCategory>
    <PaymentClass>
      <Code>PT</Code>
      <Description>Private (Independent OP)</Description>
    </PaymentClass>
  </Patient>
</Result>
```


Error Output:


```
<?xml version="1.0" encoding="UTF-8"?>
<Result>
  <Error>
    <ErrorCode>WS-00012</ErrorCode>
    <ErrorMessage>PATIENT PRN INFORMATION IS MANDATORY.</ErrorMessage>
  </Error>
</Result>
```


Remarks:

1. If passed document number has more than one patient information in VESALIUS, all the patients will be returned with the details information.
2. If returned message has a PRN empty it means person record exists but not patient.
3. Returned message has a ‘HOME’ Address as it is compulsory in VESALIUS.

Messages Used

Message Code	Message Type	Message
WS-00012	ERROR	PATIENT PRN INFORMATION IS MANDATORY.
WS-00014	ERROR	PATIENT PRN DOES NOT EXIST IN THE SYSTEM PLEASE CHECK.

3.2 GetPatientDeposit

GetPatientDeposit			
Description:	Get patient's available deposit amount.		
Input Parameters	**Description**	**Type**	**Mandatory**
PRN	Patient registration number	String	Y
COMPANY_CODE	Company code	String	Y
Sample Output:			

```
<?xml version="1.0" encoding="UTF-8"?>
<Patient>
  <Currency>BND</Currency>
  <Amount>500</Amount>
</Patient>
```

3.3 GetPatientOwingAmount

GetPatientOwingAmount			
Description:	Get patient's total owing amount.		
Input Parameters	**Description**	**Type**	**Mandatory**
PRN	Patient registration number	String	Y
COMPANY_CODE	Company code	String	Y
Sample Output:			

```
<?xml version="1.0" encoding="UTF-8"?>
<Patient>
  <Currency>BND</Currency>
  <Amount>2132</Amount>
</Patient>
```

3.4 GetPatientPastVisits

getPatientPastVisits			
Description:	Get latest N visit details and related parameters for specific Patient.		
Input Parameters	**Description**	**Type**	**Mandatory**
PRN	Patient registration number	String	Y
PAST_VISITS	Number of past visits. (Default = 1) (Max will be 10 Latest Visits)	Numeric	N
COMPANY_CODE	Company code	String	Y

Sample Output:

```
<?xml version="1.0" encoding="UTF-8"?>
<Patient>
  <Name>兩Alex Sandrea Costinha</Name>
  <PRN>F00089</PRN>
  <Visit>
    <Visit No>JP0000001711</Visit No>
    <Reg Date Time>05-Aug-2011 10:36</Reg Date Time>
    <Visit Type>OUTPATIENT</Visit Type>
    <Visit Status>OPEN</Visit Status>
    <Attending Dr>DR Srini</Attending Dr>
    <Admitting Dr></Admitting Dr>
  </Visit>
  <Visit>
    <Visit No>JP0000001701</Visit No>
    <Reg Date Time>04-Aug-2011 10:16</Reg Date Time>
    <Visit Type>OUTPATIENT</Visit Type>
    <Visit Status>OPEN</Visit Status>
    <Attending Dr>MR CHITTESH J MEHTA</Attending Dr>
    <Admitting Dr></Admitting Dr>
  </Visit>
</Patient>
```

3.5 ProcessPatientBiodata

processPatientBiodata			
Description:	To update patient information in VESALIUS.		
Input Parameters	**Description**	**Type**	**Mandatory**
TOKEN_NUMBER	Web Session ID – Used to check for active sessions only	Numeric	Y
PRN	Patient registration number (Only Valid PRN)	String	Y
DOCUMENT_TYPE	Document Type of the passed Number If multiple add with separate “**|^|**”	String	N
DOCUMENT_NO	PRN/Related Document Number	String	N

	of a Patient/Person If multiple add with separate "**|^|**" This no should match each type in previous field		
FIRST_NAME	First Name of the patient	String	Y
MIDDLE_ NAME	Middle Name of the patient	String	N
LAST_NAME	Last Name of the patient	String	N
RESIDENT	Resident of the patient Possible values are: Y – Yes, N – No (If it is empty then default value is 'N')	String	N
DOB	Date of Birth for patient (DD-MON-YYYY)	String	Y
SEX	Sex of the patient	String	Y
CONTACT	Patient Contact Number	String	Y
ADDRESS1	First line of patient's address (It is mandatory only if OPERATION_FLG = 'I')	String	Y
ADDRESS2	Second line of patient's address	String	N
ADDRESS3	Third line of patient's address	String	N
POSTAL_CODE	Postal Code of patient's address	String	N
CITY_STATE	City/State of patient's address	String	N
NATIONALITY	Nationality Code define in VESALIUS	String	Y
CHARGE_CATEGORY	Charge Category Code define in VESALIUS	String	Y
PAYMENT_CLASS	Payment Class Code define in VESALIUS	String	Y
OPERATION_FLG	Possible values are: I – Insert new patient record, U – Update existing patient record	String	Y
COMPANY_CODE	Company Code of Hospital	String	Y
WORKSTATION_CODE	Identify the valid LOCATION (Applicable only for valid WORKSTATION_CODE)	String	N
IMAGE_REF_NO	Ref No for Patient Image. INTF_WS_IN Schema (Only applicable for VESALIUS Internal Usage)	Number	N
Success Output:			
<?xml version="1.0" encoding="UTF-8"?> <Result>			

```
  <Patient>
    <PRN>2014-00001234</PRN>
    <Name>
      <Title>MS</Title>
      <FirstName>Alexandra</FirstName>
      <MiddleName></MiddleName>
      <LastName>Pheo</LastName>
    </Name>
    <Resident>Y</Resident>
    <DOB>12-Aug-1945</DOB>
    <Sex>
      <Code>F</Code>
      <Description>Female</Description>
    </Sex>
    <HomeAddress>
      <Address1>NO 18 TAMAN PUTERI JAYA</Address1>
      <Address2>JALAN 12</Address2>
      <Address3></Address3>
      <CityState>CHERAS SELANGOR</CityState>
      <PostalCode>43000</PostalCode>
    </HomeAddress>
    <ContactNumber>+65905412</ContactNumber>
    <Document>
      <Code>INTERFACE</Code>
      <Description>Interface PRN</Description>
      <Value>2014-00001234</Value>
      <ExpireDate></ExpireDate>
    </Document>
    <Document>
      <Code>PASSPORT</Code>
      <Description>Passport</Description>
      <Value>E78877998</Value>
      <ExpireDate>24-Dec-2013</ExpireDate>
    </Document>
    <Nationality>
      <Code>SG</Code>
      <Description>Singaporean</Description>
    </Nationality>
    <ChargeCategory>
      <Code>05</Code>
      <Description>NTUC MEMBER</Description>
    </ChargeCategory>
    <PaymentClass>
      <Code>PT</Code>
      <Description>Private (Independent OP)</Description>
    </PaymentClass>
  </Patient>
</Result>
```


Error Output:


```
<?xml version="1.0" encoding="UTF-8"?>
<Result>
```

```
  <Error>
    <ErrorCode>WS-00045</ErrorCode>
    <ErrorMessage>ENTER VALID OPERATION FLG VALUE.</ErrorMessage>
  </Error>
</Result>
```

Remarks:

1. After insert/update the latest information will be returned as shown in 'Success Output'.
2. For Update (OPERATION_FLG = 'U') only passed those fields which need to be updated, rest leave as empty.
3. Returned message has a 'HOME' Address as it is compulsory in VESALIUS.

Messages Used

Message Code	Message Type	Message
WS-00012	ERROR	PATIENT PRN INFORMATION IS MANDATORY.
WS-00008	ERROR	TOKEN NUMBER IS MANDATORY.
WS-00050	ERROR	OPERATION FLG IS MANDATORY.
WS-00045	ERROR	PATIENT NAME IS MANDATORY.
WS-00066	ERROR	DOB IS MANDATORY.
WS-00047	ERROR	SEX CODE IS MANDATORY.
WS-00049	ERROR	ADDRESS IS MANDATORY.
WS-00054	ERROR	NATIONALITY CODE IS MANDATORY.
WS-00055	ERROR	CHARGE CATEGORY CODE IS MANDATORY.
WS-00056	ERROR	PAYMENT CLASS CODE IS MANDATORY.
WS-00048	ERROR	INVALID SEX CODE.
WS-00009	ERROR	TOKEN NUMBER IS INVALID.
WS-00058	ERROR	INVALID NATIONALITY CODE.
WS-00059	ERROR	INVALID CHARGE CATEGORY CODE.
WS-00060	ERROR	INVALID PAYMENT CLASS CODE.
WS-00051	ERROR	INVALID OPERATION FLG.
WS-00019	ERROR	INVALID DATE FORMAT. DATE FORMAT MUST BE DD-MON-YYYY.
WS-00014	ERROR	PATIENT PRN DOES NOT EXIST IN THE SYSTEM, PLEASE CHECK. (Applicable for OPERATION_FLG= 'U')
WS-00065	ERROR	FOR INSERT (I) MODE PATIENT PRN SHOULD BE EMPTY. (Applicable for OPERATION_FLG= 'I')
WS-00062	ERROR	CONTACT IS MANDATORY.

WS-00053	ERROR	DOCUMENT TYPE IS MANDATORY.
WS-00052	ERROR	DOCUMENT NUMBER IS MANDATORY.
WS-00063	ERROR	DOCUMENT TYPE DOES NOT EXIST IN THE SYSTEM, PLEASE CHECK.
WS-00070	ERROR	INVALID VALUE FOR RESIDENT.
WS-00071	ERROR	INVALID WORKSTATION CODE.

3.6 CreatePersonToPatient

createPersonToPatient			
Description:	To Convert person record to patient record.		
Input Parameters	**Description**	**Type**	**Mandatory**
TOKEN_NUMBER	Web Session ID – Used to check for active sessions only	Numeric	Y
DOCUMENT_NUMBER	PRN/Related Document Number of a Patient/Person (Primary Document Number)	String	Y
DOCUMENT_TYPE	Document Type of the passed Number (Primary Document Type)	String	Y
OTHER_DOCUMENT_TYPE	List of other Document types, Which are separate by separator "**|^|**"	String	N
OTHER_DOCUMENT_NUMBER	List of other Document numbers (Matching Types in previous field), Which are separate by separator "**|^|**"	String	N
DOB	Date of Birth for Patient, If Person record has empty (DD-MON-YYYY)	String	Y
NATIONALITY	Nationality Code define in VESALIUS	String	Y
CHARGE_CATEGORY	Charge Category Code define in VESALIUS	String	Y
PAYMENT_CLASS	Payment Class Code define in VESALIUS	String	Y
COMPANY_CODE	Company Code for Hospital	String	Y
WORKSTATION_CODE	Identify the valid LOCATION (Applicable only for valid WORKSTATION_CODE)	String	N
Success Output:			
<?xml version="1.0" encoding="UTF-8"?>			

```
<Result>
  <Patient>
    <PRN>2014-00001234</PRN>
    <Name>
      <Title>MS</Title>
      <FirstName>Alexandra</FirstName>
      <MiddleName></MiddleName>
      <LastName>Pheo</LastName>
    </Name>
    <Resident>Y</Resident>
    <DOB>12-Aug-1945</DOB>
    <Sex>
      <Code>F</Code>
      <Description>Female</Description>
    </Sex>
    <HomeAddress>
      <Address1>NO 18 TAMAN PUTERI JAYA</Address1>
      <Address2>JALAN 12</Address2>
      <Address3></Address3>
      <CityState>CHERAS SELANGOR</CityState>
      <PostalCode>43000</PostalCode>
    </HomeAddress>
    <ContactNumber>+65905412</ContactNumber>
    <Document>
      <Code>INTERFACE</Code>
      <Description>Interface PRN</Description>
      <Value>2014-00001234</Value>
      <ExpireDate></ExpireDate>
    </Document>
    <Document>
      <Code>PASSPORT</Code>
      <Description>Passport</Description>
      <Value>E78877998</Value>
      <ExpireDate>24-Dec-2013</ExpireDate>
    </Document>
    <Nationality>
      <Code>SG</Code>
      <Description>Singaporean</Description>
    </Nationality>
    <ChargeCategory>
      <Code>05</Code>
      <Description>NTUC MEMBER</Description>
    </ChargeCategory>
    <PaymentClass>
      <Code>PT</Code>
      <Description>Private (Independent OP)</Description>
    </PaymentClass>
  </Patient>
</Result>
```


Error Output:


```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<Result>
  <Error>
    <ErrorCode>WS-00019</ErrorCode>
    <ErrorMessage>INVALID DATE FORMAT.</ErrorMessage>
  </Error>
</Result>
```

Remarks:

While creating person record to patient record pass the right pair of ‘DOCUMENT_NO’ & ‘DOCUMENT_TYPE’.

Messages Used

Message Code	Message Type	Message
WS-00052	ERROR	DOCUMENT NUMBER IS MANDATORY.
WS-00053	ERROR	DOCUMENT TYPE IS MANDATORY.
WS-00064	ERROR	DOCUMENT TYPE AND DOCUMENT NUMBER BOTH ARE MANDATORY.
WS-00008	ERROR	TOKEN NUMBER IS MANDATORY.
WS-00066	ERROR	DOB IS MANDATORY
WS-00019	ERROR	INVALID DATE FORMAT. DATE FORMAT MUST BE DD-MON-YYYY.
WS-00054	ERROR	NATIONALITY CODE IS MANDATORY.
WS-00055	ERROR	CHARGE CATEGORY CODE IS MANDATORY.
WS-00056	ERROR	PAYMENT CLASS CODE IS MANDATORY.
WS-00009	ERROR	TOKEN NUMBER IS INVALID.
WS-00058	ERROR	INVALID NATIONALITY CODE.
WS-00059	ERROR	INVALID CHARGE CATEGORY CODE.
WS-00060	ERROR	INVALID PAYMENT CLASS CODE.
WS-00063	ERROR	DOCUMENT TYPE DOES NOT EXIST IN THE SYSTEM, PLEASE CHECK.
WS-00067	ERROR	NO PERSON RECORD FOUND IN THE SYSTEM.
WS-00068	ERROR	MULTIPLE PERSON RECORDS FOUND, PLEASE UPDATE MANUALLY.
WS-00069	ERROR	PATIENT RECORD IS ALREADY EXISTS IN THE SYSTEM.
WS-00071	ERROR	INVALID WORKSTATION CODE.

3.7 ProcessPersonBiodata

ProcessPersonBiodata			
Description:	To Convert person record to patient record.		
Input Parameters	**Description**	**Type**	**Mandatory**
TOKEN_NUMBER	Web Session ID – Used to check for active sessions only	Numeric	Y
ID_Type	Primary System ID for Person (Internal usage) Not required in parameter	String	X
PERSON_ID	Person Identified, required only while calling for update mode (Operation = U)	String	N
TITLE	Person Title (Coded value as in VESALIUS)	String	N
FIRST_NAME	First Name of Person	String	Y
MIDDLE_NAME	Middle Name of Person	String	N
LAST_NAME	Last Name of Person	String	N
SEX	Gender of Person	String	Y
RESIDENT	Resident status of person Possible values = Y / N	String	Y
DOB	Date of birth in format DD-MON-YYYY	String	N
ADDRESS1	First Address Field	String	Y
ADDRESS2	Second Address Field	String	N
ADDRESS3	Third Address Field	String	N
CITY_STATE	City / State	String	N
POSTAL_CODE	Postal Code	String	N
Nationality	Nationality	String	N
CONTACT	Home Contact for Person	String	N
MARITAL_STATUS	Marital Status (coded value as defined in VESALIUS)	String	N
WORKSTATION_CODE	Identify the valid LOCATION (Applicable only for valid WORKSTATION_CODE)	String	N
OPERATION_FLAG	Insert or Update mode. I – Insert U – Update	String	Y

Success Output:

```
<?xml version="1.0" encoding="UTF-8"?>
<Result>
  <Person>
    <Name>
      <Title></Title>
      <FirstName>JOHN</FirstName>
      <MiddleName>LIEW</MiddleName>
      <LastName>TILTIS</LastName>
```

```
    </Name>
    <Resident>Yes</Resident>
    <DOB>12-Feb-1967</DOB>
    <Sex>
      <Code>M</Code>
      <Description>Male</Description>
    </Sex>
    <MaritalStatus>
      <Code></Code>
      <Description></Description>
    </MaritalStatus>
    <HomeAddress>
      <Address1>Main Street</Address1>
      <Address2>6th Lane</Address2>
      <Address3></Address3>
      <CityState>Singapore</CityState>
      <PostalCode>93933939</PostalCode>
    </HomeAddress>
    <ContactNumber>+65-209209302</ContactNumber>
    <Document>
      <Code>X2</Code>
      <Description>Person Number (Auto)</Description>
      <Value>PN17-00000013</Value>
      <ExpireDate></ExpireDate>
    </Document>
    <Nationality>
      <Code>SG</Code>
      <Description>Singaporean</Description>
    </Nationality>
  </Person>
</Result>
```


Error Output


```
<?xml version="1.0" encoding="UTF-8">
<Result>
  <Error>
    <ErrorCode>WS-00008</ErrorCode>
    <ErrorMessage>TOKEN NUMBER IS MANDATORY.</ErrorMessage>
  </Error>
</Result>
```


Message Code	Message Type	Message
WS-00008	ERROR	TOKEN NUMBER IS MANDATORY.
WS-00009	ERROR	TOKEN NUMBER IS INVALID.
WS-00005	ERROR	AUTHENTICATION FAILED.
WS-00050	ERROR	OPERATION FLG IS MANDATORY.
WS-00051	ERROR	INVALID OPERATION FLG.
WS-00071	ERROR	INVALID WORKSTATION CODE.

WS-00089	ERROR	FOR INSERT (I) MODE PERSON ID SHOULD BE EMPTY.
WS-00086	ERROR	INVALID PERSON TITLE.
WS-00090	ERROR	PERSON NAME IS MANDATORY.
WS-00047	ERROR	GENDER IS MANDATORY.
WS-00047	ERROR	INVALID GENDER CODE.
WS-00070	ERROR	INVALID VALUE FOR RESIDENT.
WS-00066	ERROR	DOB IS MANDATORY.
WS-00046	ERROR	BIRTH DATE MUST BE BEFORE CURRENT DATE.
WS-00019	ERROR	INVALID DATE FORMAT.
WS-00049	ERROR	ADDRESS IS MANDATORY.
WS-00054	ERROR	NATIONALITY CODE IS MANDATORY.
WS-00058	ERROR	INVALID NATIONALITY CODE.
WS-00062	ERROR	CONTACT IS MANDATORY.
WS-00087	ERROR	INVALID MARITAL STATUS CODE.
WS-00088	ERROR	PERSON ID IS MANDATORY.
WS-00067	ERROR	NO PERSON RECORD FOUND IN THE SYSTEM.

3.8 GetPatientQueueList

GetPatientQueueList			
Description:	Get Patient Listing by doctor for OP and IP cases, As its represented on Doctor's home page		
Input Parameters	**Description**	**Type**	**Mandatory**
TOKEN_NUMBER	Web Session ID – Used to check for active sessions only	Numeric	Y
COMPANY_CODE	Company Code	String	Y
MCR	Doctor MCR	String	Y
Type	I – Inpatient O – Outpatient Leave blank for "All"	String	N
Success Output:			

```
<?xml version="1.0" encoding="UTF-8"?>
<Result>
  <Outpatient>
    <Patient>
      <PRN>19-00000030</PRN>
      <Name>
        <Title>Mr</Title>
```

```
      <FirstName>TOZART</FirstName>
      <MiddleName>MICHAEL</MiddleName>
      <LastName>BOLONWES</LastName>
    </Name>
    <Sex>
      <Code>M</Code>
      <Description>Male</Description>
    </Sex>
    <Age>52Y 10M</Age>
    <Nationality>Singaporean</Nationality>
    <VIPFlag>NO</VIPFlag>
    <VisitType>Outpatient</VisitType>
    <VisitNumber>O-VN20-00032</VisitNumber>
    <QueueNumber>4000</QueueNumber>
    <QueueCriteria>Arrived / Inprogress</QueueCriteria>
    <PatientStatus>Inprogress</PatientStatus>
    <RegistrationDate>31-Jan-2020</RegistrationDate>
    <RegistrationTime>13:25</RegistrationTime>
    <AppointmentDate></AppointmentDate>
    <AppointmentTime></AppointmentTime>
    <VitalsAreAvailable>NO</VitalsAreAvailable>
    <TriageScore></TriageScore>
    <TriageDiscriminator></TriageDiscriminator>
    <HasOnArrivalOrders>NO</HasOnArrivalOrders>
    <RoutedBy></RoutedBy>
  </Patient>
  <Patient>
    <PRN>17-00000022</PRN>
    <Name>
      <Title>Mr</Title>
      <FirstName>TOMMY ﷺ</FirstName>
      <MiddleName>ﷺ MARTIN</MiddleName>
      <LastName>FERGES</LastName>
    </Name>
    <Sex>
      <Code>M</Code>
      <Description>Male</Description>
    </Sex>
    <Age>42Y 0M</Age>
    <Nationality>Kiwi</Nationality>
    <VIPFlag>NO</VIPFlag>
    <VisitType>Outpatient</VisitType>
    <VisitNumber>O-VN20-00031</VisitNumber>
    <QueueNumber>3000</QueueNumber>
    <QueueCriteria>Seen</QueueCriteria>
    <PatientStatus>Seen</PatientStatus>
    <RegistrationDate>31-Jan-2020</RegistrationDate>
    <RegistrationTime>11:17</RegistrationTime>
    <AppointmentDate></AppointmentDate>
    <AppointmentTime></AppointmentTime>
    <VitalsAreAvailable>NO</VitalsAreAvailable>
```

```
    <TriageScore></TriageScore>
    <TriageDiscriminator></TriageDiscriminator>
    <HasOnArrivalOrders>NO</HasOnArrivalOrders>
    <RoutedBy></RoutedBy>
  </Patient>
</Outpatient>
<Inpatient>
  <Patient>
    <PRN>19-00000020</PRN>
    <Name>
      <Title>Miss</Title>
      <FirstName>TILIPAN</FirstName>
      <MiddleName>JELONG</MiddleName>
      <LastName>MOKAITHOR</LastName>
    </Name>
    <Sex>
      <Code>F</Code>
      <Description>Female</Description>
    </Sex>
    <Age>32Y 7M</Age>
    <Nationality>Singaporean</Nationality>
    <VIPFlag>NO</VIPFlag>
    <VisitType>Inpatient</VisitType>
    <VisitNumber>IP-VN19-00044</VisitNumber>
    <Ward>TTH Main Ward 01</Ward>
    <Bed>TTH Bed 02-D01</Bed>
    <AdmissionDate>26-Dec-2019</AdmissionDate>
    <AdmissionTime>15:09</AdmissionTime>
    <TriageScore></TriageScore>
    <TriageDiscriminator></TriageDiscriminator>
  </Patient>
  <Patient>
    <PRN>17-00000062</PRN>
    <Name>
      <Title>Miss</Title>
      <FirstName>TIRLOS</FirstName>
      <MiddleName>MICHEL</MiddleName>
      <LastName>JORDEN</LastName>
    </Name>
    <Sex>
      <Code>F</Code>
      <Description>Female</Description>
    </Sex>
    <Age>41Y 7M</Age>
    <Nationality>South African</Nationality>
    <VIPFlag>NO</VIPFlag>
    <VisitType>Inpatient</VisitType>
    <VisitNumber>IP-VN20-00004</VisitNumber>
    <Ward>TTH Main Ward 01</Ward>
    <Bed>TTH Bed 01-D01</Bed>
    <AdmissionDate>14-Jan-2020</AdmissionDate>
    <AdmissionTime>13:09</AdmissionTime>
```

```
      <TriageScore></TriageScore>
      <TriageDiscriminator></TriageDiscriminator>
    </Patient>
    <Patient>
      <PRN>19-00000019</PRN>
      <Name>
        <Title>Miss</Title>
        <FirstName>TINAXI</FirstName>
        <MiddleName>CHARMIN</MiddleName>
        <LastName>SWETZ</LastName>
      </Name>
      <Sex>
        <Code>F</Code>
        <Description>Female</Description>
      </Sex>
      <Age>30Y 2M</Age>
      <Nationality>Singaporean</Nationality>
      <VIPFlag>NO</VIPFlag>
      <VisitType>Inpatient</VisitType>
      <VisitNumber>TTH-19-00000500</VisitNumber>
      <Ward>TTH Silver Garden Ward</Ward>
      <Bed>TTH Silver R1/B3</Bed>
      <AdmissionDate>10-Jul-2019</AdmissionDate>
      <AdmissionTime>13:04</AdmissionTime>
      <TriageScore></TriageScore>
      <TriageDiscriminator></TriageDiscriminator>
    </Patient>
  </Inpatient>
</Result>
```


4 APPOINTMENT

4.1 getFutureAppointment

getFutureAppointment			
Description:	To retrieve all the future appointments for the patient starting current date to the future.		
Input Parameters	**Description**	**Type**	**Mandatory**
TOKEN_NUMBER	Web Session ID – Used to check for active sessions only	Numeric	Y
COMPANY_CODE	Company code	String	Y
PRN	Patient registration number	String	Y
Success Output:			


```
<?xml version="1.0" encoding="UTF-8"?>
<Result>
  <Appointment>
```

```
    <AppointmentNumber>1110</AppointmentNumber>
    <Date>26-Apr-2012</Date>
    <Day>THURSDAY </Day>
    <StartTime>13:00</StartTime>
    <EndTime>13:05</EndTime>
    <DoctorMCR>BN236236</DoctorMCR>
    <DoctorName>Dr Srini</DoctorName>
    <SpecialityCode>ORT</SpecialityCode>
    <Speciality>ORTHOPEDIC</Speciality>
    <Clinic>Orthopedic Clinic者有</Clinic>
    <Room>OR-R1</Room>
    <CaseType>NC</CaseType>
  </Appointment>
</Result>
```

Messages Used

Message Code	Message Type	Message
WS-00015	SUCCESS	NO FUTURE APOINTMENT(S) FOUND.
WS-00008	ERROR	TOKEN NUMBER IS MANDATORY.
WS-00009	ERROR	TOKEN NUMBER IS INVALID.
WS-00012	ERROR	PATIENT PRN INFORMATION IS MANDATORY.
WS-00001	ERROR	ORGANIZATION INFORMATION IS MANDATORY.
WS-00005	ERROR	AUTHENTICATION FAILED.
WS-00013	ERROR	AUTHENTICATION FAILED PASSED TOKEN DOES NOT BELONG TO THIS ORGANIZATION.
WS-00004	ERROR	ORGANIZATION DOES NOT EXIST IN THE SYSTEM PLEASE CHECK.
WS-00014	ERROR	PATIENT PRN DOES NOT EXIST IN THE SYSTEM PLEASE CHECK.

4.2 getNextAvailableSlots

getNextAvailableSlots			
Description:	Get the next Available 5 Slots to make appointment.		
Input Parameters	**Description**	**Type**	**Mandatory**
TOKEN_NUMBER	Web Session ID – Used to check for active sessions only	Numeric	Y
COMPANY_CODE	Company code	String	Y
PRN	Patient registration number	String	Y
SPECIALTY_CODE	Specialty code to make an appointment	String	Y

MCR	Doctor MCR	String	N
START_DATE	Preferred Appointment Date In DD-MON-YYYY Format	String	Y
START_TIME	Preferred Appointment Time In HH:MM (24 Hrs Format)	String	N
CASE_TYPE	Possible values are NC – New Case, FU – Follow-up. If the external systems don't pass this value – system defaults the value to NC.	String	N

Success Output:


```
<?xml version="1.0" encoding="UTF-8"?>
<Result>
 <Slot>
  <SlotNumber>2356</SlotNumber>
  <Date>25-Apr-2012</Date>
  <Day>WEDNESDAY</Day>
  <StartTime>18:05</StartTime>
  <EndTime>18:10</EndTime>
  <DoctorName>Dr JEFFREY C. PADERNAL(放射性)</DoctorName>
  <Speciality>CARDIOLOGY</Speciality>
  <Clinic>Cardiology Clinic者有</Clinic>
  <Room>CC-R1</Room>
  <CaseType>NC</ CaseType >
 </Slot>
 <Slot>
  < SlotNumber>2357</SlotNumber>
  <Date>26-Apr-2012</Date>
  <Day>THURSDAY </Day>
  <StartTime>08:00</StartTime>
  <EndTime>08:05</EndTime>
  <DoctorName>Mr CHITTESH J MEHTA</DoctorName>
  <Speciality>CARDIOLOGY</Speciality>
  <Clinic>N/A</Clinic>
  <Room>PHY-RM1</Room>
  <CaseType>NC</CaseType>
 </Slot>
 <Slot>
  < SlotNumber >2358</SlotNumber>
  <Date>26-Apr-2012</Date>
  <Day>THURSDAY </Day>
  <StartTime>08:00</StartTime>
  <EndTime>08:05</EndTime>
  <DoctorName>Dr Kotha Sreenivas Reddy 雷迪</DoctorName>
  <Speciality>CARDIOLOGY</Speciality>
  <Clinic>Cardiology Clinic者有</Clinic>
  <Room>CC-R2</Room>
  <CaseType>NC</CaseType>
```

```
  </Slot>
  <Slot>
    <SlotNumber>2359</SlotNumber>
    <Date>26-Apr-2012</Date>
    <Day>THURSDAY </Day>
    <StartTime>13:00</StartTime>
    <EndTime>13:05</EndTime>
    <DoctorName>Mr Arvind Vasudevan</DoctorName>
    <Speciality>CARDIOLOGY</Speciality>
    <Clinic>N/A</Clinic>
    <Room>SUBB</Room>
    <CaseType>NC</CaseType>
  </Slot>
  <Slot>
    <SlotNumber>2360</SlotNumber>
    <Date>26-Apr-2012</Date>
    <Day>THURSDAY </Day>
    <StartTime>13:00</StartTime>
    <EndTime>13:05</EndTime>
    <DoctorName>Dr. Jeremy Tan</DoctorName>
    <Speciality>CARDIOLOGY</Speciality>
    <Clinic>Cardiology Clinic者有</Clinic>
    <Room>CC-R2</Room>
    <CaseType>NC</CaseType>
  </Slot>
</Result>
```


Messages Used

Message Code	Message Type	Message
WS-00024	SUCCESS	NO VALID SLOTS AVAILABLE.
WS-00008	ERROR	TOKEN NUMBER IS MANDATORY.
WS-00009	ERROR	TOKEN NUMBER IS INVALID.
WS-00012	ERROR	PATIENT PRN INFORMATION IS MANDATORY.
WS-00001	ERROR	ORGANIZATION INFORMATION IS MANDATORY.
WS-00016	ERROR	DATE INFORMATION IS MANDATORY.
WS-00017	ERROR	SPECIALTY INFORMATION IS MANDATORY.
WS-00005	ERROR	AUTHENTICATION FAILED.
WS-000013	ERROR	AUTHENTICATION FAILED PASSED TOKEN DOES NOT BELONG TO THIS ORGANIZATION.
WS-00004	ERROR	ORGANIZATION DOES NOT EXIST IN THE SYSTEM PLEASE CHECK.
WS-00014	ERROR	PATIENT PRN DOES NOT EXIST IN THE SYSTEM PLEASE CHECK.
WS-00018	ERROR	GIVEN DATE IS PAST DATE.

WS-00019	ERROR	INVALID DATE FORMAT.
WS-00020	ERROR	SPECIALTY DOES NOT EXIST IN THE SYSTEM PLEASE CHECK.
WS-00021	ERROR	GIVEN TIME IS PAST TIME.
WS-00022	ERROR	INVALID TIME FORMAT.
WS-00023	ERROR	MCR DOES NOT EXIST IN THE SYSTEM PLEASE CHECK.
WS-00007	ERROR	DATABASE ERROR.

4.3 MakeAppointment

makeAppointment			
Description:	To make a new appointment		
Input Parameters	**Description**	**Type**	**Mandatory**
TOKEN_NUMBER	Web Session ID – Used to check for active sessions only	Numeric	Y
COMPANY_CODE	Company code	String	Y
PRN	Patient registration number	String	Y
SLOT_NUMBER	Selected Slot number	Numeric	Y
CASE_TYPE	Possible values are NC – New Case, FU – Follow-up. If the external systems don't pass this value – system defaults the value to NC.	String	N

Success Output:

```
<?xml version="1.0" encoding="UTF-8"?>
<Result>
  <AppointmentBookingConfirmation>
    <AppointmentStatus>CONFIRMED</AppointmentStatus>
    <AppointmentNumber>1113</AppointmentNumber>
    <Date>26-Apr-2012</Date>
    <Day>THURSDAY </Day>
    <StartTime>13:00</StartTime>
    <EndTime>13:05</EndTime>
    <DoctorName>Dr Jeremy Tan</DoctorName>
    <Speciality>CARDIOLOGY</Speciality>
    <Clinic>Cardiology Clinic者有</Clinic>
    <Room>CC-R2</Room>
    <CaseType>NC</CaseType>
  </AppointmentBookingConfirmation>
</Result>
```

Messages Used

Message Code	Message Type	Message
WS-00008	ERROR	TOKEN NUMBER IS MANDATORY.
WS-00009	ERROR	TOKEN NUMBER IS INVALID.
WS-00012	ERROR	PATIENT PRN INFORMATION IS MANDATORY.
WS-00001	ERROR	ORGANIZATION INFORMATION IS MANDATORY.
WS-00025	ERROR	SLOT NUMBER IS MANDATORY.
WS-00026	ERROR	SLOT NUMBER IS INVALID.
WS-00027	ERROR	APPOINTMENT TYPE MUST BE EITHER NEW CASE OR FOLLOW UP.
WS-00005	ERROR	AUTHENTICATION FAILED.
WS-00013	ERROR	AUTHENTICATION FAILED PASSED TOKEN DOES NOT BELONG TO THIS ORGANIZATION.
WS-00004	ERROR	ORGANIZATION DOES NOT EXIST IN THE SYSTEM PLEASE CHECK.
WS-00014	ERROR	PATIENT PRN DOES NOT EXIST IN THE SYSTEM PLEASE CHECK.
WS-00028	ERROR	SLOT NUMBER DOES NOT EXIST IN THE SYSTEM PLEASE CHECK.
WS-00029	ERROR	SLOT NUMBER AND PATIENT PRN MISMATCH PLEASE CHECK.
WS-00007	ERROR	DATABASE ERROR.
WS-00030	ERROR	SPECIFIED SLOT NUMBER DOES NOT BELONG TO ORGANIZATION PLEASE CHECK.

4.4 ChangeAppointment

changeAppointment			
Description:	To change an existing appointment		
Input Parameters	**Description**	**Type**	**Mandatory**
TOKEN_NUMBER	Web Session ID – Used to check for active sessions only	Numeric	Y
COMPANY_CODE	Company code	String	Y
PRN	Patient registration number	String	Y
SLOT_NUMBER	Selected Slot number	Numeric	Y
APPOINTMENT_NUMBER	The appointment which is to be changed	Numeric	Y
REASON	The reason for the change in appointment	String	Y

Success Output:


```
<?xml version="1.0" encoding="UTF-8"?>
<Result>
  <AppointmentChangeConfirmation>
    <AppointmentStatus>CHANGED</AppointmentStatus>
    <AppointmentNumber>1111</AppointmentNumber>
    <Date>26-Apr-2012</Date>
    <Day>THURSDAY </Day>
    <StartTime>08:00</StartTime>
    <EndTime>08:05</EndTime>
    <DoctorName>Dr Kotha Sreenivas Reddy 雷迪</DoctorName>
    <Speciality>CARDIOLOGY</Speciality>
    <Clinic>Cardiology Clinic者有</Clinic>
    <Room>CC-R2</Room>
    <CaseType>FU</CaseType>
  </AppointmentChangeConfirmation>
</Result>
```


Messages Used

Message Code	Message Type	Message
WS-00008	ERROR	TOKEN NUMBER IS MANDATORY.
WS-00009	ERROR	TOKEN NUMBER IS INVALID.
WS-00001	ERROR	ORGANIZATION INFORMATION IS MANDATORY.
WS-00012	ERROR	PATIENT PRN INFORMATION IS MANDATORY.
WS-00031	ERROR	APPOINTMENT NUMBER IS MANDATORY.
WS-00032	ERROR	APPOINTMENT NUMBER FORMAT IS INVALID.
WS-00025	ERROR	SLOT NUMBER IS MANDATORY.
WS-00026	ERROR	SLOT NUMBER IS INVALID.
WS-00033	ERROR	REASON IS MANDATORY.
WS-00038	ERROR	REASON CAN NOT BE MORE THAN 200 CHARACTERS.
WS-00005	ERROR	AUTHENTICATION FAILED.
WS-00004	ERROR	ORGANIZATION DOES NOT EXIST IN THE SYSTEM PLEASE CHECK.
WS-00013	ERROR	AUTHENTICATION FAILED PASSED TOKEN DOES NOT BELONG TO THIS ORGANIZATION.
WS-00014	ERROR	PATIENT PRN DOES NOT EXIST IN THE SYSTEM PLEASE CHECK.
WS-00034	ERROR	APPOINTMENT NUMBER DOES NOT EXIST IN THE SYSTEM PLEASE CHECK.

WS-00035	ERROR	APPOINTMENT NUMBER IS NOT VALID FOR THIS PATIENT PRN PLEASE CHECK.
WS-00028	ERROR	SLOT NUMBER DOES NOT EXIST IN THE SYSTEM PLEASE CHECK.
WS-00029	ERROR	SLOT NUMBER AND PATIENT PRN MISMATCH PLEASE CHECK.
WS-00036	ERROR	PATIENT HAVE AN APPOINTMENT FOR THE SELECTED DATE AND TIME.
WS-00037	ERROR	APPOINTMENT CANNOT BE CHANGED TO ANOTHER DOCTOR.
WS-00007	ERROR	DATABASE ERROR.
WS-00030	ERROR	SPECIFIED SLOT NUMBER DOES NOT BELONG TO ORGANIZATION PLEASE CHECK.
WS-00041	ERROR	PAST APPOINTMENTS CAN NOT BE MODIFIED.

4.5 CancelAppointment

cancelAppointment			
Description:	To cancel an existing appointment		
Input Parameters	**Description**	**Type**	**Mandatory**
TOKEN	Web Session ID – Used to check for active sessions only	Numeric	Y
COMPANY_CODE	Company code	String	Y
PRN	Patient registration number	String	Y
APPOINTMENT_NUMBER	The appointment which is to be changed	Numeric	Y
REASON	The reason for cancellation	String	Y

Success Output:

```
<?xml version="1.0" encoding="UTF-8"?>
<Result>
  <AppointmentCancelConfirmation>
    <AppointmentStatus>CANCELLED</AppointmentStatus>
    <AppointmentNumber>48679</AppointmentNumber>
    <Date>16-Jul-2012</Date>
    <Day>MONDAY</Day>
    <StartTime>12:00</StartTime>
    <EndTime>12:10</EndTime>
    <DoctorName>DR. Srini</DoctorName>
    <Specialty>ORTHOPEDI</Specialty>
    <Clinic>N/A</Clinic>
    <Room>INSTALL</Room>
    <CaseType>NC</CaseType>
  </AppointmentCancelConfirmation>
</Result>
```

Messages Used

Message Code	Message Type	Message
WS-00008	ERROR	TOKEN NUMBER IS MANDATORY.
WS-00009	ERROR	TOKEN NUMBER IS INVALID.
WS-00001	ERROR	ORGANIZATION INFORMATION IS MANDATORY.
WS-00012	ERROR	PATIENT PRN INFORMATION IS MANDATORY.
WS-00031	ERROR	APPOINTMENT NUMBER IS MANDATORY.
WS-00032	ERROR	APPOINTMENT NUMBER FORMAT IS INVALID.
WS-00033	ERROR	REASON IS MANDATORY.
WS-00038	ERROR	REASON CAN NOT BE MORE THAN 200 CHARACTERS.
WS-00005	ERROR	AUTHENTICATION FAILED.
WS-00004	ERROR	ORGANIZATION DOES NOT EXIST IN THE SYSTEM PLEASE CHECK.
WS-00013	ERROR	AUTHENTICATION FAILED PASSED TOKEN DOES NOT BELONG TO THIS ORGANIZATION.
WS-00014	ERROR	PATIENT PRN DOES NOT EXIST IN THE SYSTEM PLEASE CHECK.
WS-00007	ERROR	DATABASE ERROR.
WS-00035	ERROR	APPOINTMENT NUMBER IS NOT VALID FOR THIS PATIENT PRN PLEASE CHECK.
WS-00041	ERROR	PAST APPOINTMENTS CAN NOT BE MODIFIED.

4.6 GetDoctorData

GetDoctorData			
Description:	Retrieve all valid (Active) doctor details from VESALIUS with basic doctor details.		
Input Parameters	**Description**	**Type**	**Mandatory**
TOKEN	Web Session ID – Used to check for active sessions only	Numeric	Y
COMPANY_CODE	Company code	String	Y
Success Output:			
<Doctor> <MCR>23929229</MCR> <Name>			

```
    <Title>Mr</Title>
    <FirstName>ALEX</FirstName>
    <MiddleName>Prince</MiddleName>
    <LastName>CHARLES MARTIN</LastName>
  </Name>
  <ShortName>ACM</ShortName>
  <Grade>VISITING CONSULTANT (客座顧問)</Grade>
  <DOB>02-Feb-1956</DOB>
  <Gender>
    <Code>M</Code>
    <Description>Male</Description>
  </Gender>
  <Address>
    <AddressType>HOME</AddressType>
    <Address1>123</Address1>
    <Address2>Leviing Road</Address2>
    <Address3></Address3>
    <Address4>Singapore</Address4>
    <Address5>989889</Address5>
  </Address>
  <Contact>123123123</Contact>
  <Document>
    <Code>X2</Code>
    <Description>Person Number (Auto)</Description>
    <Value>PN13-00000035</Value>
    <ExpiryDate></ExpiryDate>
  </Document>
  <Nationality>
    <Code>SG</Code>
    <Description>Singaporean</Description>
  </Nationality>
  <Specialty>
    <Code>ONC</Code>
    <Description>ONCOLOGY</Description>
    <Primary>YES</Primary>
  </Specialty>
  <Image>
    <Type></Type>
    <Height></Height>
    <Width></Width>
    <Base64ImageData></Base64ImageData>
  </Image>
</Doctor>
```


4.7 GetSpecialtyData

GetDoctorData

Description:	Retrieve all valid Specialty defined in VESALISUS.		
Input Parameters	**Description**	**Type**	**Mandatory**
TOKEN	Web Session ID – Used to check for active sessions only	Numeric	Y
COMPANY_CODE	Company code	String	Y
Success Output:			

```
<?xml version="1.0" encoding="UTF-8"?>
<Result>
  <Specialty>
    <Code>01</Code>
    <Description>Cardiology</Description>
  </Specialty>
  <Specialty>
    <Code>02</Code>
    <Description>Neurology</Description>
  </Specialty>
  <Specialty>
    <Code>ONC</Code>
    <Description>Oncology</Description>
  </Specialty>
  <Specialty>
    <Code>03</Code>
    <Description>Orthopedic</Description>
  </Specialty>
</Result>
```

4.8 GetDoctorAvailabilityData

GetDoctorData			
Description:	Retrieve all available slots for a specified Doctor for one week from the date passed.		
Input Parameters	**Description**	**Type**	**Mandatory**
TOKEN	Web Session ID – Used to check for active sessions only	Numeric	Y
COMPANY_CODE	Company code	String	Y
MCR	Doctor MCR	String	Y
Date	Start Date Data will be retrieved for 7 days, Beginning from this date Required format : DD-MON-YYYY a.e 01-JAN-2017	String	Y
Success Output:			

```
<?xml version="1.0" encoding="UTF-8"?>
<Result>
  <Availability>
```

```
<28-Jul-2017>
  <Session>
    <StartTime>13:00</StartTime>
    <EndTime>14:00</EndTime>
  </Session>
  <Session>
    <StartTime>14:20</StartTime>
    <EndTime>15:00</EndTime>
  </Session>
  <Session>
    <StartTime>15:10</StartTime>
    <EndTime>16:00</EndTime>
  </Session>
  <Session>
    <StartTime>16:15</StartTime>
    <EndTime>17:00</EndTime>
  </Session>
  <Session>
    <StartTime>17:20</StartTime>
    <EndTime>18:00</EndTime>
  </Session>
</28-Jul-2017>
<29-Jul-2017>
  <Session>
    <StartTime></StartTime>
    <EndTime></EndTime>
  </Session>
</29-Jul-2017>
<30-Jul-2017>
  <Session>
    <StartTime></StartTime>
    <EndTime></EndTime>
  </Session>
</30-Jul-2017>
<31-Jul-2017>
  <Session>
    <StartTime></StartTime>
    <EndTime></EndTime>
  </Session>
</31-Jul-2017>
<01-Aug-2017>
  <Session>
    <StartTime>10:00</StartTime>
    <EndTime>12:00</EndTime>
  </Session>
  <Session>
    <StartTime>14:30</StartTime>
    <EndTime>18:00</EndTime>
  </Session>
</01-Aug-2017>
<02-Aug-2017>
  <Session>
```

```
      <StartTime></StartTime>
      <EndTime></EndTime>
    </Session>
  </02-Aug-2017>
  <03-Aug-2017>
    <Session>
      <StartTime>14:00</StartTime>
      <EndTime>15:00</EndTime>
    </Session>
    <Session>
      <StartTime>15:20</StartTime>
      <EndTime>18:00</EndTime>
    </Session>
  </03-Aug-2017>
 </Availability>
</Result>
```


4.9 ProcessAppointmentBookingFee

ProcessAppointmentBookingFee			
Description:	This WS will allow to collect Appointment Deposit fee based on Appointment made in VESLAIUS using Appointment Booking Web Services.		
Input Parameters	**Description**	**Type**	**Mandatory**
TOKEN_NUMBER	Web Session ID – Used to check for active sessions only	Numeric	Y
PRN	Patient PRN	String	Y
APPOINTMENT_REF_NO	ARN, Which is provided by Appointment Booking WS once appointment is confirmed.	String	Y
AMOUNT	Amount to be collected as Appointment Booking Fee	Number	Y
COMPANY_CODE	Company code	String	Y
PAYMENT_CODE	Code for Payment Mode to be used for collection as defined in VESALIUS	String	Y
Success Output:			


```
<?xml version="1.0" encoding="UTF-8"?>
<Appointment>
 <ReceiptNo>17-006781</ReceiptNo>
 <ReceiptDateTime>01-Dec-2017 14:00</ReceiptDateTime>
 <Amount>20.00</Amount>
</Appointment>
```


5 CLINICAL

5.1 ProcessRadiologyReport

Process PDF Radiology Report			
Description:	Updating the PDF document for incoming interface data for PDF (Base64) radiology report.		
Input Parameters	**Description**	**Type**	**Mandatory**
TOKEN	Web Session ID – Used to check for active sessions only	Numeric	Y
COMPANY_CODE	Company code	String	Y
PRN	Patient registration number	String	Y
Account No.	Patient Visit Number	String	Y
Order ID	Radiology Order Number	Number	Y
Event Seq No	Event Sequence Number, Referred from Incoming VESALIUS Interface	Number	Y
Success Output:			
<?xml version="1.0" encoding="UTF-8"> <Result> <Success> <Code>WS-00099</Code> <Message>RADIOLOGY REPORT PDF UPLOAD WAS SUCCESSFUL.</Message> </Success> </Result>			

5.2 ProcessLaboratorytReport

Process PDF Laboratory Report			
Description:	Updating the PDF document for incoming interface data for PDF (Base64) laboratory report.		
Input Parameters	**Description**	**Type**	**Mandatory**
TOKEN	Web Session ID – Used to check for active sessions only	Numeric	Y
COMPANY_CODE	Company code	String	Y
PRN	Patient registration number	String	Y
Account No.	Patient Visit Number	String	Y
Order ID	Laboratory Order Number	Number	Y
Event Seq No	Event Sequence Number, Referred from Incoming VESALIUS Interface	Number	Y
Success Output:			
<?xml version="1.0" encoding="UTF-8"> <Result> <Success>			

```
    <Code>WS-00099</Code>
    <Message>LAB  REPORT PDF UPLOAD WAS SUCCESSFUL.</Message>
  </Success>
</Result>
```


5.3 GetInvestigationReport

Retrieve all investigation results (Ready for Review)			
Description:	This web service will retrieve all Investigation results which are ready for review as presented on Doctor's homepage.		
Input Parameters	**Description**	**Type**	**Mandatory**
TOKEN	Web Session ID – Used to check for active sessions only	Numeric	Y
COMPANY_CODE	Company code	String	Y
INVESTIGATION_TYPE	LABS – Laboratory orders XRAY – Xray Orders DI – Diagnostic Investigation ALL – All of above	String	Y
ACCOUNT_NO	Account No, If its required to retrieve all orders for a specific account No	String	N
PRN	Patient PRN	String	N
DATE	Order date for retrieving all orders (This is mandatory, If Account No is not passed) Date should be in DD-MON-YYYY Format.	String	N
Success Output:			


```
<?xml version="1.0" encoding="UTF-8"?>
<Result>
  <Investigation>
    <Visit>
      <AccountNo>O-VN20-00031</AccountNo>
      <PRN>17-00000022</PRN>
      <Patient>Mr TOMMY FERGES</Patient>
      <Reports>
        <Service>
          <AccessionNo>10462</AccessionNo>
          <Type>LABS</Type>
          <Code>LS0026337</Code>
          <Description>DGP</Description>
          <OrderDoctorMcr>ZB67890</OrderDoctorMcr>
          <OrderDate>31-Jan-2020</OrderDate>
          <ResultDate>31-Jan-2020</ResultDate>
          <ReportType>PDF</ReportType>
          <Result>Base64 Content for the PDF Report</Result>
          <Remark></Remark>
        </Service>
        <Service>
          <AccessionNo>10461</AccessionNo>
```

```
        <Type>LABS</Type>
        <Code>ELPH-X3-L45</Code>
        <Description>ELECTROPHOROSYS SEGMENT TESTING (Z-3 Method) :
คำอธิบายยาวแล็บ โรงพยาบาลไทย - รุ่นสุดท้าย</Description>
        <OrderDoctorMcr>ZB67890</OrderDoctorMcr>
        <OrderDate>31-Jan-2020</OrderDate>
        <ResultDate>31-Jan-2020</ResultDate>
        <ReportType>MANUAL</ReportType>
        <Result>* LAB PANEL *</Result>
        <Remark></Remark>
      </Service>
      <Service>
        <AccessionNo>10452</AccessionNo>
        <Type>LABS</Type>
        <Code>HDL</Code>
        <Description>CHOLESTROL - HDL</Description>
        <OrderDoctorMcr>CJ007</OrderDoctorMcr>
        <OrderDate>31-Jan-2020</OrderDate>
        <ResultDate>31-Jan-2020</ResultDate>
        <ReportType>MANUAL</ReportType>
        <Result>345.00</Result>
        <Remark></Remark>
      </Service>
      <Service>
        <AccessionNo>10450</AccessionNo>
        <Type>LABS</Type>
        <Code>LSVB12</Code>
        <Description>VITAMIN B12</Description>
        <OrderDoctorMcr>CJ007</OrderDoctorMcr>
        <OrderDate>31-Jan-2020</OrderDate>
        <ResultDate>31-Jan-2020</ResultDate>
        <ReportType>MANUAL</ReportType>
        <Result>24:13</Result>
        <Remark></Remark>
      </Service>
    </Reports>
  </Visit>
  <Visit>
    <AccountNo>O-VN20-00032</AccountNo>
    <PRN>19-00000030</PRN>
    <Patient>Mr TOZART BOLONWES</Patient>
    <Reports>
      <Service>
        <AccessionNo>10467</AccessionNo>
        <Type>DI</Type>
        <Code>EEG</Code>
        <Description>EEG (脑电图)</Description>
        <OrderDoctorMcr>CJ007</OrderDoctorMcr>
        <OrderDate>31-Jan-2020</OrderDate>
        <ResultDate>31-Jan-2020</ResultDate>
        <ReportType>MANUAL</ReportType>
```

```
        <Result>DI Test Result</Result>
        <Remark></Remark>
      </Service>
    </Reports>
  </Visit>
 </Investigation>
</Result>
```

5.4 ProcessReviewInvestigation

Update Review status in VESALIUS for external Review			
Description:	This web service will update Investigation review with date and time for ordered doctor.		
Input Parameters	**Description**	**Type**	**Mandatory**
TOKEN	Web Session ID – Used to check for active sessions only	Numeric	Y
COMPANY_CODE	Company code	String	Y
PRN	Patient PRN	String	Y
ACCESSION_NO	Unique Identified of the Investigation, which is only provided by the calling WebService "GetInvestigationReport"	String	Y
REVIEW_DOC	Review Doctor MCR (This should be ordering doctor)	String	Y
REVIEW_DATE	Review Date in DD-MON-YYYY Format	String	Y
REVIEW_TIME	Review Time is 24hrs Format(HH:MM)	String	Y
Success Output:			

```
<?xml version="1.0" encoding="UTF-8">
<Result>
 <Success>
  <Code>WS-00149</Code>
  <Message>INVESTIGATION REPORT IS REVIEWED
SUCCESSFULLY.</Message>
 </Success>
</Result>
```

5.5 GetDoctorToDoNotification

Doctor ToDo notification for following types, Which are appearing in VESALIUS Homepage
- Verbal Order
- Drug Discontinue
- Discharge Summary Notification (Up to 72 Hrs of Discharge)

Description:	This web service will update Investigation review with date and time for ordered doctor.		
Input Parameters	**Description**	**Type**	**Mandatory**
TOKEN	Web Session ID – Used to check for active sessions only	Numeric	Y
COMPANY_CODE	Company code	String	Y
NOTIFICATION	Filter for selecting only specific notification. Default : All OV : Verbal Order DV : Discontinue Drug DSN : Discharge Summary	String	N
DOCTOR_MCR	MCR code for Doctor	String	Y

Success Output:


```
<?xml version="1.0" encoding="UTF-8"?>
<Result>
  <Notification>
    <PRN>19-00000114</PRN>
    <PatientName>PIXESH DHRAMCHAND WADHWA, Mr</PatientName>
    <AccountNo>IP-VN19-00043</AccountNo>
    <DoctorName>Mr CHITTESH J MEHTA</DoctorName>
    <VerbalOrder>
      <Item>
        <AccessionNo>2269</AccessionNo>
        <Code>ANAREXT500I</Code>
        <Description>ANAREX TABLETS 500 MG (PARACETAMOL + ORPHENADRINE
CITRATE 35MG)</Description>
        <Quantity>6</Quantity>
        <UOM>TABLET</UOM>
        <Instruction>Take 1 Tablets Three Times Daily</Instruction>
        <OrderDate>18-Feb-2020</OrderDate>
        <OrderTime>15:09</OrderTime>
        <OrderedBy>NATALIE</OrderedBy>
        <DispenseQty></DispenseQty>
        <LastDispenseTime></LastDispenseTime>
        <ServedQty></ServedQty>
        <LastServedTime></LastServedTime>
      </Item>
    </VerbalOrder>
  </Notification>
  <Notification>
    <PRN>19-00000116</PRN>
    <PatientName>PINAMOL CHARLES MORTON, Mr</PatientName>
    <AccountNo>IP-VN19-00046</AccountNo>
    <DoctorName>Mr CHITTESH J MEHTA</DoctorName>
    <DischargeSummary>Outstanding</DischargesSummary>
  </Notification>
  <Notification>
    <PRN>20-00000008</PRN>
    <PatientName>PIYUSH CHANDRAMOHAN GAMBHELKAR</PatientName>
```

```
    <AccountNo>IP-VN20-00007</AccountNo>
    <DoctorName>Mr CHITTESH J MEHTA</DoctorName>
    <DrugDiscontinue>
      <Item>
        <AccessionNo>2265</AccessionNo>
        <Code>ANAREXT500I</Code>
        <Description>ANAREX TABLETS 500 MG (PARACETAMOL + ORPHENADRINE
CITRATE 35MG)</Description>
        <Quantity>0</Quantity>
        <UOM>TABLET</UOM>
        <Instruction>Take 1 Tablets Two Times Daily</Instruction>
        <OrderDate>18-Feb-2020</OrderDate>
        <OrderTime>10:51</OrderTime>
        <OrderedBy>CHITTESH</OrderedBy>
        <DiscontinueDate>18-Feb-2020</DiscontinueDate>
        <DiscontinueTime>15:11</DiscontinueTime>
        <DiscontinueReason>Patient Request</DiscontinueReason>
        <DiscontinueBy>NATALIE</DiscontinueBy>
      </Item>
    </DrugDiscontinue>
  </Notification>
</Result>
```


5.6 ProcessDoctorToDoAcknowledge

Doctor ToDo notification Acknowledgement for particular Notification			
Description:	This web service will update Investigation review with date and time for ordered doctor.		
Input Parameters	**Description**	**Type**	**Mandatory**
TOKEN	Web Session ID – Used to check for active sessions only	Numeric	Y
COMPANY_CODE	Company code	String	Y
TYPE	Notification ACK type VOA : Verbal Order ACK VOAD : Verbal Order Discontinue DISCONTINUE-ACK : Drug Order Discontinue ACK	String	Y
ACCESSION_NO	No of particular Notification, retrieved while accessing notifications	String	Y
DOCTOR_MCR	Doctor's MCR	String	Y
REMARKS	Remark for Acknowledgement	String	N
Success Output:			


```
<?xml version="1.0" encoding="UTF-8">
<Result>
  <Success>
    <Code>WS-00158</Code>
```

```
    <Message>ACKNOWLEDGED SUCCESSFULLY.</Message>
  </Success>
</Result>
```

6 ADT

6.1 GetProformaBill

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Get Proforma Bill Amount for Patient			
Description:	This Web Service will return outstanding Bill Amount, Which requested patient is due and will appear at the Billing functions in VESALIUS		
Input Parameters	**Description**	**Type**	**Mandatory**
TOKEN	Web Session ID – Used to check for active sessions only	Numeric	Y
COMPANY_CODE	Company code	String	Y
PRN	Patient registration number	String	Y
Success Output:			
<?xml version="1.0" encoding="UTF-8"?> <Result> <Patient> <PRN>PRN126376</PRN> <VisitNumber>HOSP-Y18-00001</VisitNumber> <Amount>32650.00</Amount> </Patient> </Result>			

6.2 ProcessBedStatus

Process Bed Status to Available			
Description:	Process Bed Status to available after housekeeping work is done, Following discharge in VESALIUS		
Input Parameters	**Description**	**Type**	**Mandatory**
TOKEN	Web Session ID – Used to check for active sessions only	Numeric	Y
COMPANY_CODE	Company code	String	Y
WARD_NO	Ward No	String	Y
BED_NO	Bed No	String	Y
Success Output:			
<?xml version="1.0" encoding="UTF-8"?> <Result> <Success> <WardNo>WARD-01</WardNo> <BedNo>BED-W1-B1</BedNo> <Status>BED STATUS HAS BEEN CHANGED TO AVAILABLE.</Status>			

</Success> </Result>			